Ludwig Alvarado Becerra

Systems Engineer | Computational Modeling & Simulation
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PROFILE

Systems Engineering and Computational Modeling & Simulation professional awaiting graduation, with 2+ years of hands-on experience in software automation, CI/CD, GNU/Linux systems, containerization, and application reliability. Professional experience in software automation at ROSEN, complemented by technical infrastructure and automation work as a founding member of a geotechnology organization. Experienced with C#, Python, Bash, GitHub Actions, Docker, Kubernetes, and Linux environments. Interested in Site Reliability Engineering, production automation, observability, and reliable software delivery.

EXPERIENCE

Automation Engineer Intern Jan. 2026 – Aug. 2026
ROSEN - RTRC Bogotá *Bogotá, Colombia*

- Automated software validation workflows for Oil & Gas pipeline inspection applications, contributing to an estimated USD 1.5M in savings over four months.
- Improved CI/CD build performance by 30% and modernized legacy application modules in C#.

Founding Member – Technical Infrastructure Nov. 2024 – Present
Asociación de Cartografía Colaborativa de Colombia (AC3) *Bogotá, Colombia*

- Maintain GNU/Linux environments and automate system configuration with Bash, while troubleshooting networking and application environments.
- Containerize and maintain geospatial workflows using Docker, providing reproducible infrastructure for drone-based data acquisition and processing.

EDUCATION

Systems Engineering – Computational Modeling & Simulation Jan. 2022 – Nov. 2026
Universidad Jorge Tadeo Lozano *Bogotá, Colombia*

- Founder and coordinator of Segmentation Fault, a competitive programming group focused on algorithms and problem solving for .
- Founder of GNUTADEO, a free software community focused on GNU/Linux, open-source technologies, and technical knowledge sharing.

RECOGNITIONS

1st Place – RiskTech Hackathon Nov. 2025
RiskTech *Bogotá, Colombia*

- Designed a reproducible software workflow for fraud detection in banking transactions using machine learning.

2nd Place – Hackathon Desafío Inteligente Dec. 2025
Ministry of Mines and Energy – Universidad Tecnológica de Pereira *Bogotá, Colombia*

- Developed a digital-twin software solution for monitoring reservoir levels in Colombia, emphasizing reproducible and open-source software workflows.

TECHNICAL SKILLS

Systems & Infrastructure: GNU/Linux, Bash, Docker, Podman, Kubernetes, Networking Fundamentals.
CI/CD & Automation: CI/CD, GitHub Actions, Git, Playwright, Python, Shell Scripting.
Development: C#, Python, C++, Java, SQL, JavaScript, TypeScript, REST APIs.
Engineering: Software Testing, Application Reliability, Troubleshooting, Automation, Distributed Systems.
Languages: Spanish (Native), English (B2), Mandarin Chinese (HSK3).